

Soil plasticity and strength – a new approach using extrusion

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THE MOST CONSPICUOUS property of a clay soil as its water content changes is its variation in consistency in which it can be observed that as water content increases, strength decreases. A series of limits to consistency were first suggested by Atterberg, of which the liquid limit, W_L , plastic limit, W_p , and shrinkage limit, W_s , have been adopted by the construction industry. These are defined as arbitrary water contents at which a soil becomes sufficiently weak so as to flow like a fluid (i.e. W_L) or sufficiently stiff so that it becomes brittle and fractures at failure (i.e. W_p). The shrinkage limit, W_s , represents the onset of partial saturation during the drying process from a soil previously in a saturated remoulded condition (Fig. 1).

It has been shown (Schofield & Wroth, 1968) that for saturated remoulded soils the liquid and plastic limits define points on the critical state line such that the loss of water content, or plasticity index PI, between these limits corresponds to an arbitrary proportional increase of strength. The plasticity index also provides a measure of the soil's plastic compressibility.

The plasticity limits, therefore, can be considered as water contents associated with states of strength of a soil in a remoulded saturated condition. Tests to establish these limits will directly or indirectly be strength tests, and will have a reliability dependent upon the accuracy upon which the strength is determined. It is proposed in this note that the arbitrary strengths that can be associated with the liquid and plastic limits, as traditionally determined by British Standards, are

at liquid limit, $c_u = 1.6 \text{ kN/m}^2$

at plastic limit, $c_u = 110 \text{ kN/m}^2$

It is proposed that the water contents of saturated remoulded soil relevant to these strengths be defined as the liquid and plastic limits, and suggested that suitable techniques to determine these are triaxial testing, cone penetrometers and soil extrusion tests.

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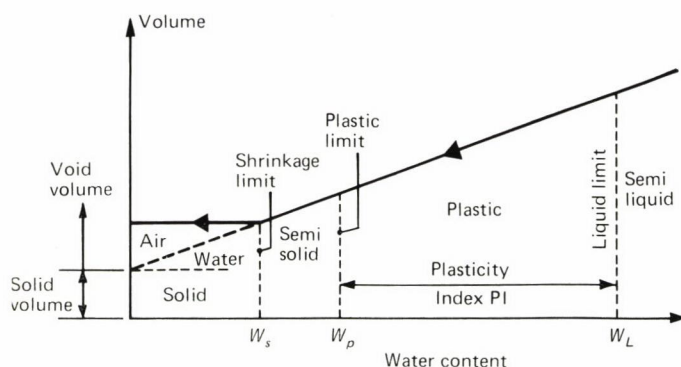


Fig. 1. Consistency of soil

The liquid limit

Atterberg's original procedure was developed into a percussion cup mechanical device (Fig. 2) to standardise test procedures and to attempt to measure indirectly a constant value of shearing resistance at the liquid limit (Casagrande 1932, 1958). In this test, the liquid limit is defined as the water content at which a 2mm wide groove closes over a base distance of 13mm under 25 standard blows. This water content, however, is influenced by:

- material of the base,
- base dimensions and its insulation from the supporting table,
- material, dimensions and weight of the cup,
- vertical drop height of the cup,
- soil type, and
- operator judgement.

In particular, item (a) has varied in different countries as shown in Table I, and tests (Norman, 1958) have showed that the softer BS 1377 bases produced higher liquid limit values than the harder Ameri-

can type bases. For example, results on London Clay were

$$W_L = 0.72 \text{ (BS 1377)}$$

$$W_L = 0.68 \text{ (Micarta base)}$$

Table I shows that current UK practice still adopts a 'soft' base and thus reported liquid limit values will be wetter, and weaker, than those from American type tests. Norman (1958) showed the 'hard' base strength at the liquid limit to be about 25% higher, and care is required in defining a value of strength to be assigned to the liquid limit. Karlsson (1977) and Wroth (1979) have also demonstrated the liquid limit strength from percussion cup tests to vary with soil type. Published values of shear strength, determined by laboratory vane tests, are given in Table II.

The percussion cup test has been proved to have poor reproducibility (Sherwood, 1970) and this perhaps is not surprising considering the variations in operator judgement, machine design and strength variability with soil type. A much more

TABLE I. LIQUID LIMIT PERCUSSION TEST BASES

Base material	Hardness	Resilience	Source
Ebonite	—	—	Casagrande (1932)
Micarta 221	98.5°	—	Norman (1958)
Rubber	86° – 89°	—	BS 1377 (1948)
4 rubber laminations	86° – 89°	30% – 40%	BS 1377 (1961)
4 rubber laminations	86° – 94°	20% – 35%	BS 1377 (1963)
4 vulcanised rubber laminations	86° – 94°	20% – 35%	BS 1377 (1967–1975)

TABLE II. SHEAR STRENGTH AT LIQUID LIMIT

Base material	c_u (kNm ²)	Source
Ebonite	2.5	Seed <i>et al</i> (1964)
Micarta	2 – 3	Casagrande (1958)
	1.1 – 2.3	Norman (1958)
	1.3 – 2.4	Youssef <i>et al</i> (1965)
	0.5 – 4.0	Karlsson (1977)
BS Rubber	0.8 – 1.6	Norman (1958)
	0.7 – 1.45	Skempton & Northey (1952)
	1 – 3	Skopek & Ter-Stepanian (1975)

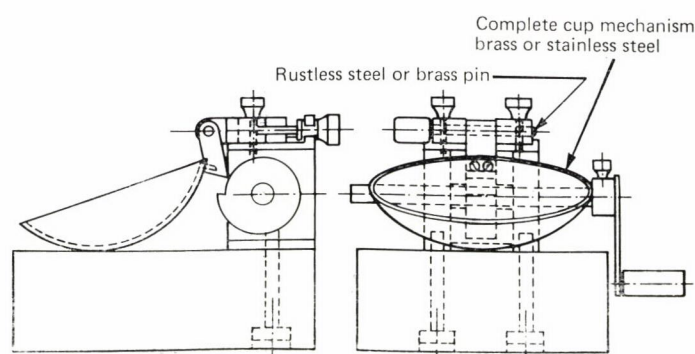


Fig. 2. Percussion cup liquid limit apparatus

(after Casagrande 1958 and BS 1377:1975)

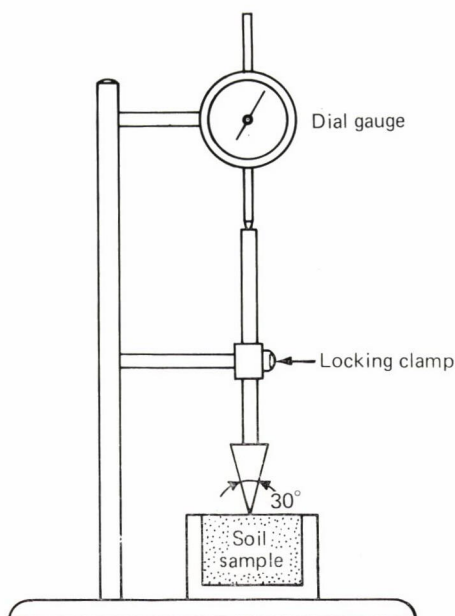


Fig. 3. BS cone penetration test

reliable and reproducible indirect measure of soil strength at the liquid limit is provided by cone testing.

Cone penetration tests originated in Sweden in 1915, and Fig. 3 outlines the test details. A review of tests (Sherwood & Ryley, 1968) recommended for UK usage an 80g cone of 30° angle and defined the water content at 20mm penetration as being the liquid limit. It has been shown (Hansbo, 1957; Wroth & Wood, 1978; Karlsson, 1961) that for a particular cone

$$\frac{c_u d^2}{W} = \text{a constant, } k$$

where d = cone penetration
 W = cone weight

and $k \approx 0.80$ for 30° cones
 $k \approx 0.27$ for 60° cones

In Sweden, a 60g cone of 60° angle and 10mm penetration has been adopted for the liquid limit test, and Karlsson (1977) has shown this represents a shear strength of about 1.7kN/m² for all soil types, with the results correlating well with past prac-

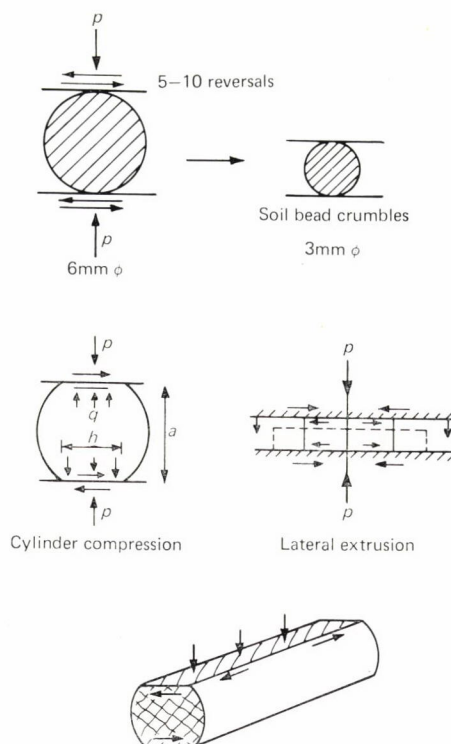


Fig. 4. Plastic limit test

tice using hard-base percussion machines. The BS 1377 (1975) 80g cone also correlates well with soft-base percussion machines, and the above implies the strength at the liquid limit to be about 1.6kN/m², a value slightly smaller than that for the Swedish cone. This is to be expected since the cones were correlated against different standards, but suggests that in practice the value for W_L obtained from cone tests would be similar and fortuitously result in a reasonably consistent international standard. The inference on liquid limit values is indicated in Table III.

The plastic limit

The test for determining the onset of brittleness of a soil has remained relatively unchanged since Terzaghi (1926a) modified Atterberg's procedure by specifying a diameter of 3mm to the soil bead that crumbles on rolling, the water content at this condition being defined as the

plastic limit. This procedure subjects the soil to a very complex stress system in that it combines bar rolling distortion, cylinder compression and lateral extrusion processes (Fig. 4). A rigorous analysis for these stresses does not exist but, assuming full saturation and incompressibility, plasticity theory indicates the soil yield stress to be functions of:

- (i) applied pressure to the soil bead,
- (ii) geometry, e.g. width of hand contact (h) to bead diameter (a), i.e. h/a ratio
- (iii) friction between soil, hand and base plate, and
- (iv) speed of rolling.

None of these variables are controlled in the rolling bead test, and the known variation of results (Sherwood, 1970) is not surprising. For example, a clay with mean plastic limit $W_p = 0.23$ when tested in different laboratories produced results ranging from as low as 0.19 to as high as 0.39. Reported values of shear strength at the plastic limit (Table IV) cover a large range from 20–320kN/m² which at first sight appears to invalidate a strength criterion as a basis for the limit. This range of strength, however, is no worse than that previously obtained with the percussion cup liquid limit test (i.e. 0.5–4.0kN/m²), allowing for the exponential form of strength-water content relationships. The rolling bead test, therefore, cannot be expected to provide reliable and consistent results for plastic limit, since the shear strength is inadequately controlled and dependent on many variables inherent to the test procedure. It is of interest to note that doubts about the Atterberg tests for soil consistency were first made by Terzaghi (1926b), when it was suggested that cube strengths from compression tests could provide a better basis.

The advantages of simple test method and low cost for the rolling bead plastic limit are more than outweighed by the complexity of the mechanical process and the crudity of the results obtained. Many attempts to improve on this have been proposed, e.g. 'squeeze' tests (Burmister, 1936; Warlam, 1936), soil suction (Upal, 1966; Livneh *et al.*, 1970; Gedalleh *et al.*, 1974; Russell & Mickle, 1970), consolidation (Nuyens & Kockaerts, 1967), lineal shrinkage (Woods, 1960), cone penetrometers (Wroth & Wood, 1978) and extrusion (Timar, 1974). None of these methods have as yet been generally adopted as a suitable replacement to the long established bead test, but it is considered that cone penetrometers and extrusion techniques offer the most promising developments for the future since these are based on more reliable determinations of shear strength. The question to be answered, however, is what value of strength can be arbitrarily assigned to denote the onset of brittleness in a soil?

The undrained strength of a saturated remoulded soil is known to be a function of many parameters such as method of test, strain rate, etc. In order to produce a standard, therefore, it is proposed that the conventional triaxial compression test should form the basis for comparison, as also proposed by Wroth (1979). Table IV gives reported shear strengths that indicate an average value of 110–130kN/m² at the plastic limit water content. On the assumption that an average value reflects the 'true' plastic limit of a material, then it is suggested that the strength best associated with the first crumble point of a soil bead is the lower value, i.e. 110kN/

TABLE III. LIQUID LIMITS FOR SOILS (after Norman, 1958)

Soil	BS Percussion Test		BS Cone ($c_u = 1.6 \text{ kN/m}^2$)	Swedish Cone ($c_u = 1.7 \text{ kN/m}^2$)
	W_L	$c_u \text{ kN/m}^2$	W_L	W_L
Brickearth	0.41	1.4	0.40	0.39
Weald Clay	0.48	1.6	0.48	0.48
London Clay	0.72	1.1	0.66	0.65
Oxford Clay	0.67	0.8	0.59	0.59
Gault Clay	0.72	1.6	0.72	0.71

(Cone W_L results estimated from strength values)

TABLE IV. SHEAR STRENGTH AT PLASTIC LIMIT, W_p

Source	$c_u \text{ kN/m}^2$ Range	Average
Skempton & Northey (1952)	85 – 125	110
Dennehy (1978)	30 – 320	115 (arithmetic) 104 (geometric)
Arrowsmith (1978)	20 – 220	110
Arrowsmith (1979)	25 – 280	130
Wroth (1979)	170 (assumed)	

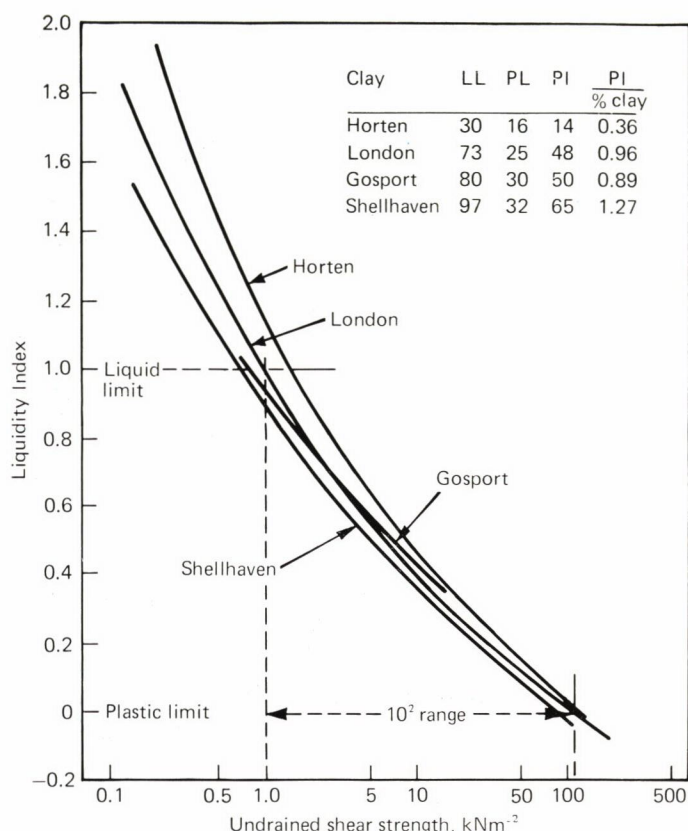


Fig. 5. Relation between shear strength and liquidity index (after Skempton & Northey, 1952)

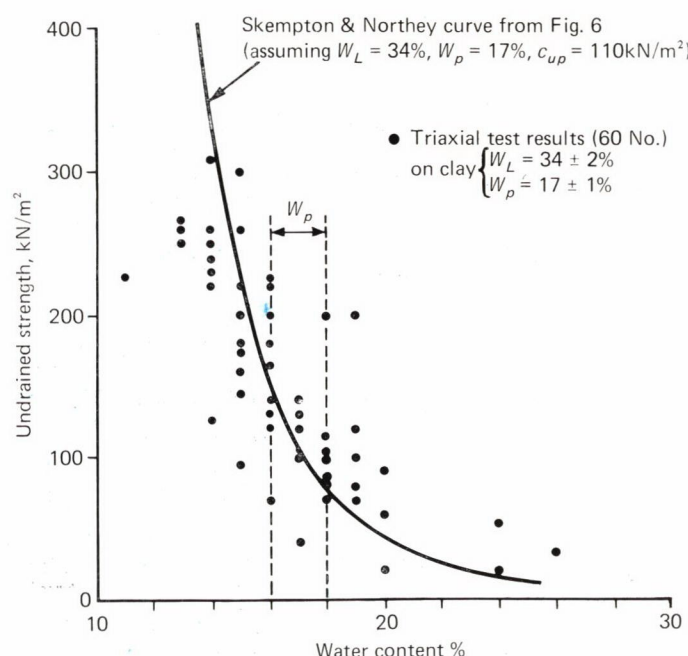


Fig. 7. Undrained strength water content plot from commercial tests on a sandy clay

m². This also represents a consistency state about mid-way in the 'stiff' range for soils.

Data on strength and water content has been provided by Skempton & Northey (1952), as shown in Fig. 5. In this, the water contents have been normalised into liquidity index (or consistency index) where

$$\text{Liquidity Index, LI} = \frac{W - W_p}{W_L - W_p}$$

$$\text{Consistency Index, CI} = \frac{W_L - W}{W_L - W_p}$$

The liquid limits were determined by 'soft' base percussion machines and plastic limits by rolling beads, and the data has been used (Schofield & Wroth, 1968) to define a strength range for soil plasticity of:

$$\frac{c_{up}}{c_{uL}} = 100$$

where c_{up} = strength at plastic limit, W_p
 c_{uL} = strength at liquid limit, W_L

If however, the liquid limit is now defined as the water content associated with a strength of 1.6 kN/m² (as inferred from

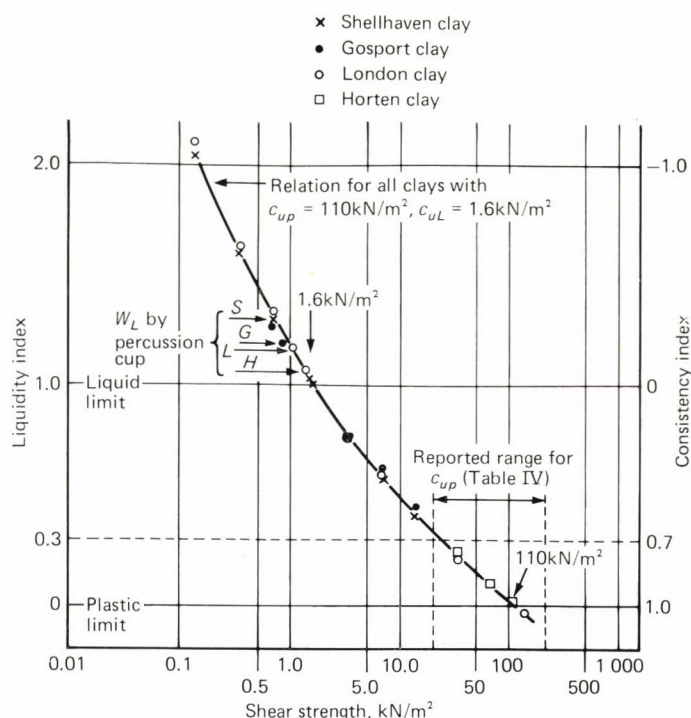


Fig. 6. Relation between liquidity index (consistency index) and shear strength of remould clays (after Skempton & Northey, 1952)

the BS cone test) and the plastic limit as the water content associated with a strength of 110 kN/m², then the strength ratio c_{up}/c_{uL} becomes about 70 and the data in Fig. 5 can be replotted as shown in Fig. 6. This yields an apparently unique, almost log linear, relationship between strength and consistency, as predicted by critical state theory, and produces the following correlations:

$$PI = (W_L - W_p) \approx 0.59 (W_L - 0.09)$$

$$C_c' \approx 1.45 PI$$

$$c_u \approx 1.6 \exp (4.23 CI)$$

where C_c' is the compression index of the soil.

From the above, it is evident that for a low plasticity soil (say $PI = 0.10$), then a $\pm 1\%$ error in determining water content produces a $\pm 42\%$ error in shear strength. As the rolling bead test can produce errors of at least $\pm 3\%$ water content for plastic limit, the wide range of shear strengths reported in Table IV are not to be unexpected, given the crudity of the test.

Fig. 7 shows test data from a site investigation (Arrowsmith, 1979) where plasticity tests were carried out on each triaxial sample (650 tests in total) and shows results from soils of similar reported plasticity. The usual 'measles' plot is apparent, but a curve based on Skempton & Northey's data and assigning strengths of 1.6 kN/m² and 110 kN/m² to the reported plasticity limits does provide a good fit to the results, indicating these arbitrarily chosen strengths to be of the right order. Further tests at UMIST on remoulded low plasticity glacial clay statically compacted to near saturation are shown in Fig. 8, and again these are compared with the curve obtained from Fig. 6. The plasticity properties for this clay are:

$$W_L = 0.325 \text{ (cone penetrometer)}$$

$$W_p = 0.148 - 0.172 \text{ (rolling bead)}$$

$$= 0.162 \text{ average (S.D. = 0.7)}$$

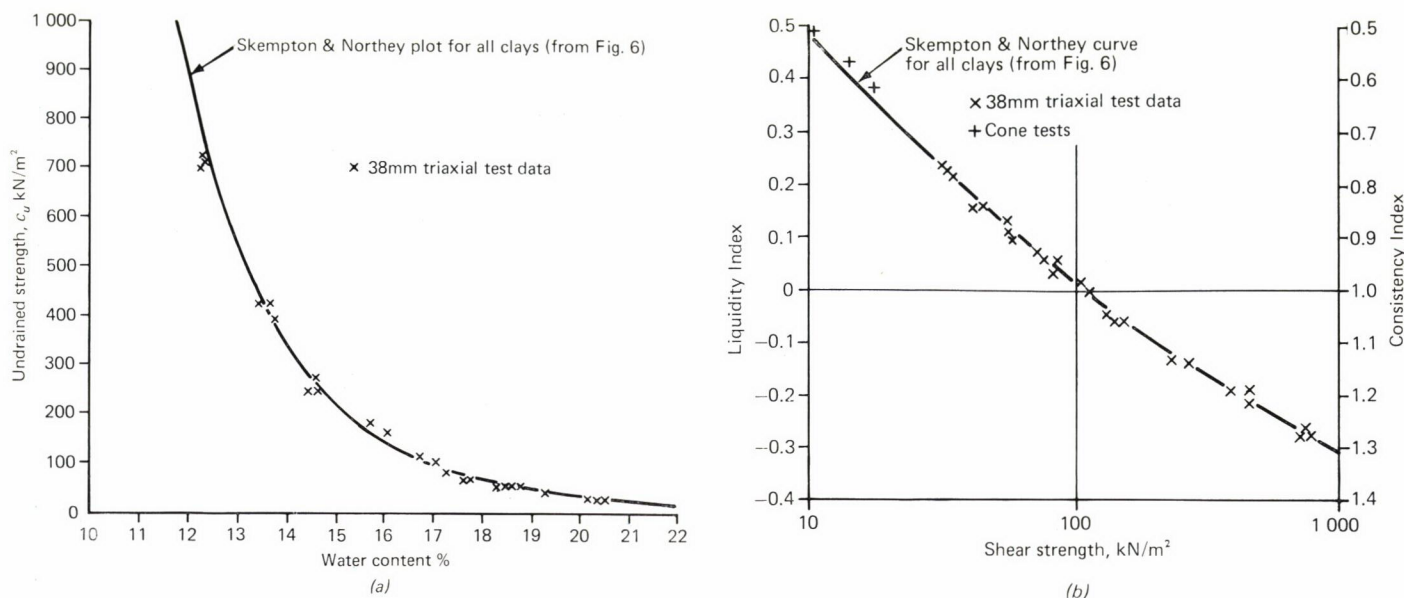


Fig. 8 (a) Undrained strength water content for remoulded sandy clay. (b) Consistency index shear strength for remoulded sandy clay

The value of W_p inferred from a strength of $110kN/m^2$ is:

$$W_p = 0.165$$

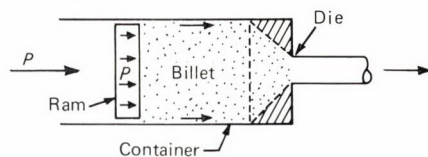
The above evidence suggests, therefore, that if plasticity is defined on a strength basis then there exists an apparently unique strength-water content relationship for saturated remoulded clays. Soil strength would form a more rational and reliable method of test for plasticity and the only questions to be decided are what values of strength should be arbitrarily adopted to be reasonably consistent with past practice. It is suggested that these should be:

$$\begin{aligned} c_{uL} &= 1.6kN/m^2 \text{ (BS cone test)} \\ c_{uL} &= 1.7kN/m^2 \text{ (Swedish cone test)} \\ c_{up} &= 110kN/m^2 \end{aligned}$$

A value for c_{up} of $170kN/m^2$ has previously been assumed by Wroth (1979) based on the data of Youssef *et al* (1965) and the 100-fold strength range reported by Skempton & Northey (1952). The liquid limits of Youssef *et al*, however, were from 'hard' base machines, and thus do not correlate with the 'soft' machine used in Britain in 1952. The convenient multiplier of 100 for strength over the plastic range cannot now be adopted since cone tests are based on slightly stronger soil at the liquid limit. If a value of $170kN/m^2$ is arbitrarily chosen to represent strength at the plastic limit, then a result is obtained which is on average lower than from a rolling bead test, the amount of the variation increasing with increasing plasticity index as shown in Table V.

The major problem in determining plastic limit from shear strength tests relates

Direct extrusion



Reverse extrusion

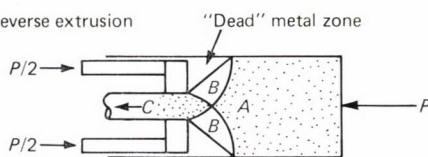


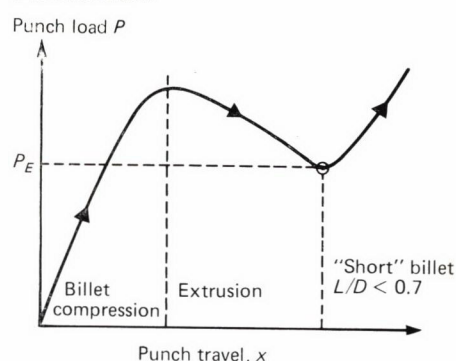
Fig. 9. Direct and reverse extrusion

to test technique. Soil at the plastic limit is stiff and difficult to mix in a saturated remoulded condition, and inconsistent results arise from samples that are partially saturated. To determine plastic limit, therefore, from a strength criterion it is necessary to test either statically compacted soils at varying water contents, or dynamically compacted samples at high cell pressures to entrain air into solution, and then establish the strength-water content relationship. This procedure is cumbersome compared with rolling bead tests, but investigations in progress indicate that an extrusion process may provide a simple, cheap and economic alternative.

Soil extrusion

Extrusion as a mechanical process was probably first developed at the end of the 18th century for forming lead pipes, and applied to soils for brick manufacturing purposes since the mid-19th century. There are two methods for extruding materials,

Direct extrusion



Reverse extrusion

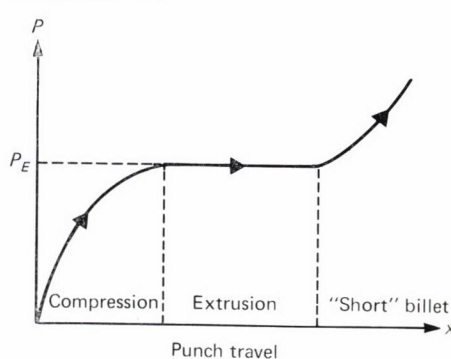


Fig. 10. Extrusion punch loads

direct or reverse extrusion, as shown in Fig. 9, producing typical punch load/travel diagrams shown in Fig. 10. The extrusion pressure, P , is defined as:

$$P = \frac{P_E}{A}$$

where P_E = steady state extrusion load, and

A = container area

The relationship between extrusion pressure and reduction in billet area has been shown (Johnson & Mellor, 1962; Johnson & Kudo, 1962) to be well fitted by the empirical equation:

$$\frac{P}{k} = a + b \ln \frac{A}{A_0}$$

TABLE V. COMPARISON OF PLASTIC LIMITS

Soil	Plastic Limit		
	Rolling bead	$c_u = 110kN/m^2$	$c_u = 170kN/m^2$
London	0.25	0.25	0.22
Gosport	0.30	—	—
Shellhaven	0.32	0.29	0.25
Horten	0.16	0.16	0.15
Glacial (1)	0.162	0.165	0.156
Glacial (2)	0.17±0.1	0.17	0.16

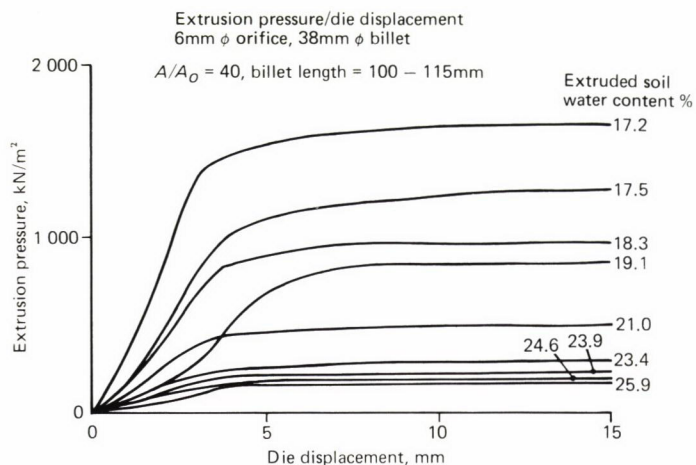


Fig. 11. Typical test results on "long" billets

where A_0 = area of die orifice,
 k = yield stress in shear,
 a, b are constants, with
 $a = 1.0 - 1.6$
 $b = 3.0 - 3.6$

In reverse extrusion (Fig. 9) of metal, the billet A is rigidly contained and the die advances with a certain velocity. When the metal reaches the shear zone B , it distorts and leaves the die as a rigid bar C . Each element of metal is rigid until it enters the fan, where it shears and flows under the severe and local effect of the die. The material is stressed to its yield limit, and it is not possible to transmit more pressure from the billet to the die face through this yielding plastic zone, unless the zone of shear alters from that shown in Fig. 9 such as when the die punch approaches the end of the container to form a short billet ($L/D < 0.75$). This formation of plastic yield, if present in soils, suggests the test to be of a suitable form for determining plasticity such that a

relationship of the following form may be predicted:

$$\frac{P}{c_u} = a' + b' \ln \frac{A}{A_0}$$

where a', b' are constants.

Figs. 11 and 12 show typical results for extrusion pressures on a low plasticity clay ($W_L = 0.325, W_p = 0.165$) where reverse extrusion was applied with a container 38mm diameter and die orifice 6mm diameter ($A/A_0 = 40$). Similar tests at other area ratios produced the results shown in Fig. 13, giving a best fit result of:

$$\frac{P}{c_u} = 1.6 + 4.3 \ln \frac{A}{A_0}$$

These results, however, are to be considered only as an indication of the stress ratio since the equipment used was relatively crude. A more accurate extrusion system is presently being developed for

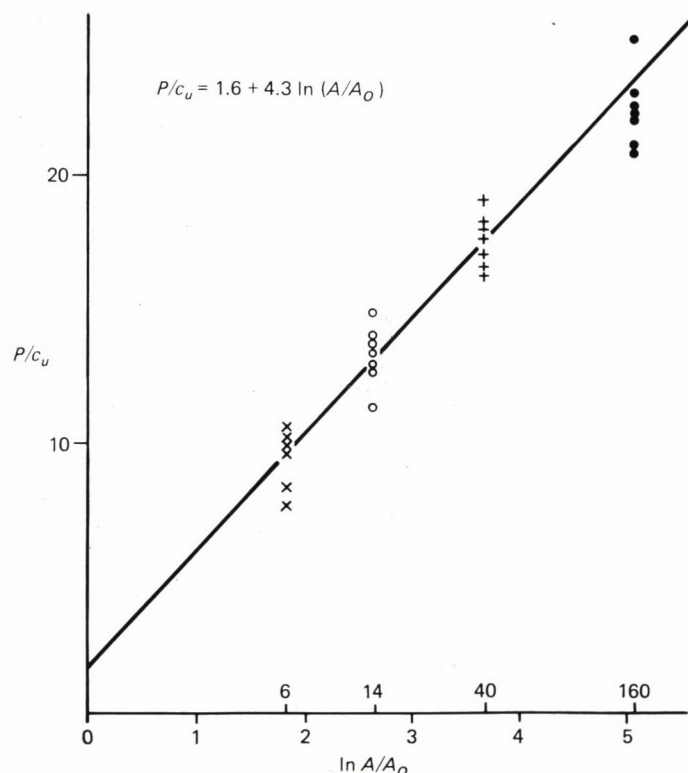


Fig. 13. Extrusion pressure ratio and area reduction

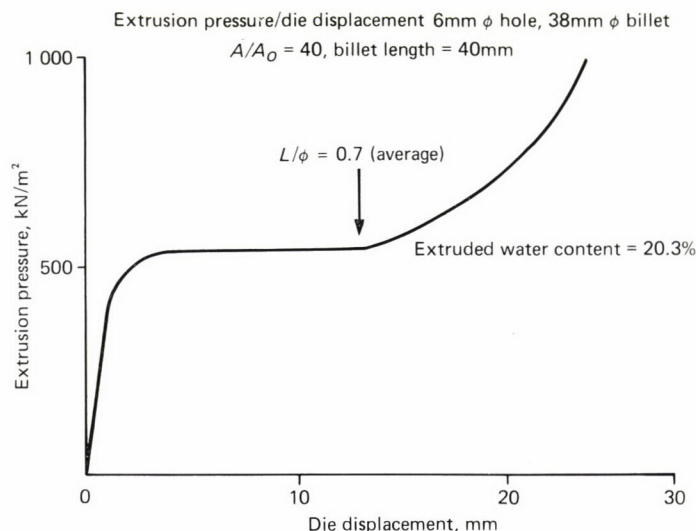


Fig. 12. Typical test result on a "short" billet

commercial usage and this will enable the above relationship to be more reliably defined for soils. Nevertheless, the above results using crude equipment are sufficiently encouraging, and similar to the plasticity testing of other materials, to justify a more serious examination of the technique. This is further supported by Figs. 14 and 15 where the extrusion pressure/water content relationships for the area ratios tested are presented, and compared to predicted extrusion pressure curves obtained using the stress ratio P/c_u given above and the shear strength/water content values obtained from the Skempton & Northey data re-examined in Fig. 6.

From Figs. 14 and 15, it can be seen that even with the admittedly crude equipment used for this preliminary test series, there is a very close correlation between extrusion pressure P and shear strength c_u that is of the exponential form required. Thus knowing the P/c_u ratio for a particular area ratio, and measuring the extrusion pressure, the shear strength of a soil as determined in a triaxial test can be reliably obtained. Further evidence of this form of relationship is provided from the brick industry (Clews, 1969) where experiments have also shown the existence of a "yield value, Y " applying to plastic flow where

$$Y = A e^{-a w}$$

where w = water content, and
 A, a are constants.

Direct extrusion tests on soils (Timar, 1974) have also shown the validity of the technique for determining both plastic and liquid limits, though the method of interpretation is more difficult due to the influence of friction as the billet is forced along the container to the die.

Extrusion can be used therefore as a method to establish the strength/water content relationship for a remoulded soil and thus establish the water content associated with a strength of 110 kN/m^2 , i.e. the plastic limit, and if required can be extended down to the liquid limit as an alternative procedure to cone penetration. The benefits obtained from extrusion can be summarised as being:

1. A small soil sample only is required for testing (c. 200g in the present test). The sample can be progressively wetted, mixed and extruded.
2. The test equipment is simple, robust and probably inexpensive to produce on a production basis.

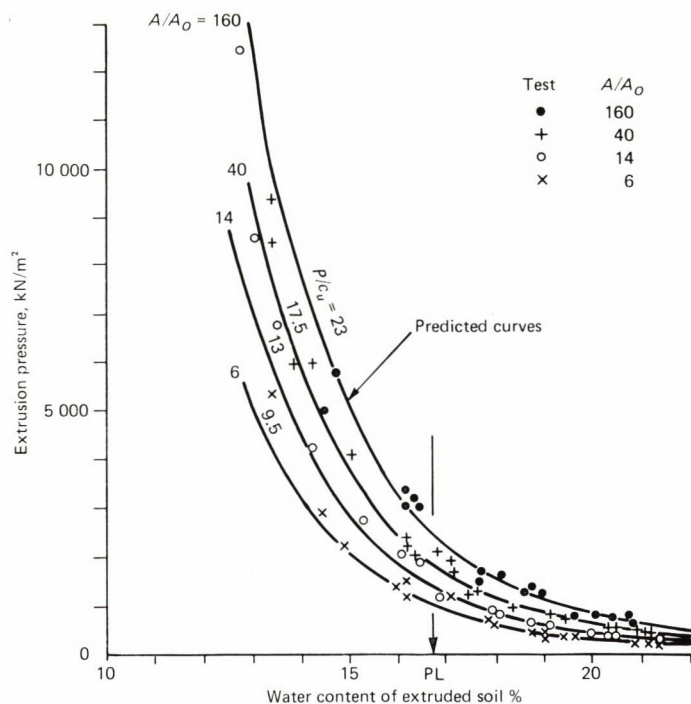


Fig. 14. Extrusion pressure and water content

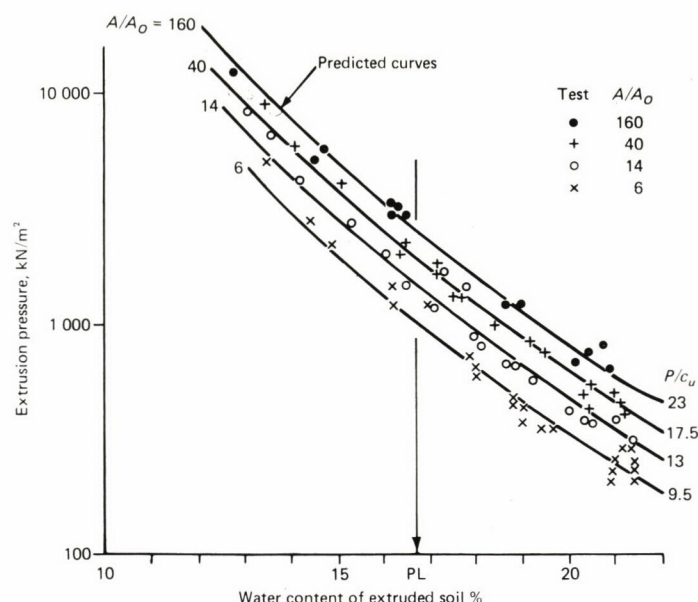


Fig. 15. Extrusion pressure and soil water content

3. The remoulded soil is rigidly contained in the container and subjected to relatively high pressures such that any air voids would be driven into solution with the pore water. It has been confirmed in the preliminary test series that full saturation is effectively achieved.

4. Testing soil under saturated conditions produces results of a greater reliability and reproducibility. Alternative simple strength tests such as laboratory vanes, cones, penetrometers and unconfined shear are less reliable due to the difficulties of remoulding to saturation stiff pastes of soil.

5. The extrusion pressure correlates well with undrained triaxial strength tests.

6. If the extruded soil worm is chosen to have a diameter of 6mm, then a quick 'rolling bead' test can always be carried out as a check on the results if required (see BS 1377, 1975 in which a 6mm bead is specified).

7. The test is relatively rapid to carry out, and should not take significantly longer than present methods. It should therefore present an economic alternative of greater reliability.

Conclusions

(a) The consistency of soils can be represented by the liquid and plastic limits, which are water contents of saturated remoulded clays that define critical states for the soil.

(b) The plastic and liquid limit critical states are related to unique shear strengths, the proposed conditions being:

$$\begin{aligned}\text{strength at liquid limit} &= 1.6\text{kN/m}^2 \\ \text{strength at plastic limit} &= 110\text{kN/m}^2\end{aligned}$$

(c) The strength at the liquid limit is reliably determined by the cone penetration test conducted to the standards of BS 1377 (1975). The Casagrande type percussion cup machine is not reliable and should not form a basis of test.

(d) The strength at the plastic limit is not reliably established by the Atterberg type rolling bead test and this is not a suitable method.

(e) Extrusion testing is a well estab-

lished technique in both metal-forming and brick-making which is dependent upon plastic yield of the material.

(f) Preliminary reverse extrusion tests on a low plasticity clay produced results favourable with other plastic materials, and enabled shear strengths to be determined with a high reliability.

(g) Extrusion of soils is a reliable method for determining soil plasticity that promises to be rapid, simple and economical. It is also based on well-established mechanical principles.

(h) It is to be emphasised that the extrusion data presented are of a preliminary nature that have to be confirmed by future tests on test equipment currently being developed.

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